

LOGICWorld: Automated Deep Logic Generation for Interactive Reasoning in Embodied Agents

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Abstract—The successful deployment of household robots in physical environments relies on their ability to comprehend and execute complex, conditional human instructions. In reality, human commands are rarely static or linear; they often involve dynamic decision-making, such as, “If there is a Coke in the fridge, bring it to me; otherwise, go to the kitchen and turn off the light.” Executing such tasks requires the robot to possess robust interactive reasoning and the capacity for logical state-machine tracking. However, current embodied AI benchmarks predominantly feature simplified action spaces and sequential task compositions, failing to adequately assess the dynamic reasoning and non-linear planning cognitive frameworks required for real-world robotics. In this study, we introduce LOGICWorld, a novel cognitive framework and benchmark specifically designed to evaluate the interactive reasoning competence of embodied agents. LOGICWorld significantly increases task complexity by employing an automated mechanism to synthesize conditional instructions governed by intricate propositional logic. This design mandates dynamic goal evaluation, active environmental exploration, and non-linear conditional branching, forcing the robotic agent to actively update its internal state machine based on sensory feedback and historical facts. Our evaluation of state-of-the-art LLM-driven cognitive engines reveals a critical vulnerability: while agents achieve over 90% success rates in basic spatial execution, their performance severely degrades to below 65% in the Compositional Reasoning category. Through this rigorous construction, LOGICWorld provides a challenging and necessary standard for evaluating the true logical and cognitive readiness of next-generation household robots.

I. INTRODUCTION

The ultimate goal of Embodied Intelligence is to deploy household robots capable of seamlessly assisting humans in unstructured, open-world physical environments. A critical prerequisite for such autonomy is the robot’s ability to interpret and execute natural, complex human spoken instructions. In everyday scenarios, human commands are rarely simple, step-by-step directives. Instead, they frequently encapsulate conditional logic, uncertainty, and dynamic priorities—for example, “If there is a Coke in the fridge, bring it to me; otherwise, go to the kitchen and turn off the light.” To successfully fulfill such instructions, a robot cannot rely on a pre-computed, linear trajectory. It must act as an intelligent, dynamic agent capable of robust **logical state-machine tracking**: actively exploring the environment (e.g., opening the fridge), updating its internal belief state based

on partial observations, evaluating logical conditions, and branching its execution plan accordingly.

The recent paradigm shift enabled by Large Language Models (LLMs) and Vision-Language Models (VLMs) has provided a highly promising cognitive engine, or “brain,” for these embodied agents. The emergence of Large Reasoning Models (LRMs) such as OpenAI’s o1 [1] and DeepSeek-R1 [2] has demonstrated impressive zero-shot generalization and step-by-step reasoning. Consequently, researchers have rapidly adopted LLMs to generate hierarchical plans and low-level execution sequences for robotic control [3]–[5]. While these LLM-driven cognitive frameworks excel at semantic grounding, their reliability in tracking complex, multi-tiered logical states during long-horizon physical execution remains an open question.

We posit that existing embodied interactive reasoning benchmarks fail to adequately stress-test this crucial cognitive dimension. Prior evaluation tasks [6], [7] have heavily focused on straightforward, static manipulation (e.g., ‘put the lettuce in the refrigerator’). While newer benchmarks have expanded the physical scale and multimodal diversity of environments [8], [9], their underlying task structures still largely map to sequential goal lists or simple conjunctions. This persistent simplicity overlooks the necessity for dynamic self-correction and non-linear plan adjustment. Consequently, current benchmarks function more as basic tests of physical grounding rather than rigorous evaluations of a robot’s high-level logical reasoning and cognitive flexibility.

To address this critical gap in embodied robotics, we introduce **LOGICWorld**, a comprehensive cognitive evaluation framework designed to rigorously assess the logical state-machine tracking capabilities of household robots. Rather than merely scaling up the physical environment, we systematically push the boundaries of robotic cognition by employing an automated mechanism to generate tasks based on **intricate, dynamic propositional logic**. This approach bridges the rigorous, structured state representations of classical planning (e.g., PDDL) with the open-world semantic capabilities required by modern LLM-driven agents.

We categorize robotic tasks into three hierarchical levels of increasing cognitive demand: (1) **Spatial Execution**, focusing on foundational long-horizon navigation and sequential interaction; (2) **State Tracking**, introducing deterministic constraints, working memory maintenance, and basic conditional branching; and (3) **Compositional Reasoning**, mandating deep propositional inference through complex, multi-branch logical expressions (e.g., ‘ $A \wedge (B \vee C)$ ’) that mimic the most convoluted human spoken instructions.

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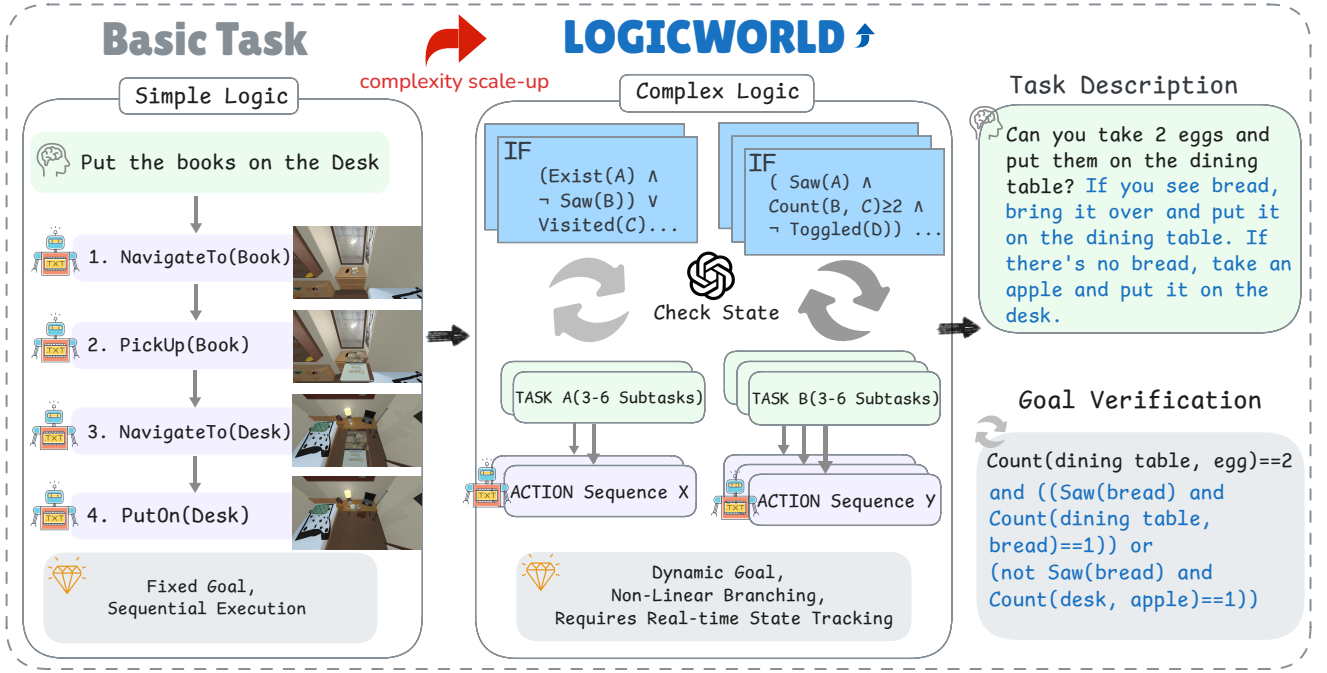


Fig. 1. Overview of LOGICWorld. **Left:** Depicts static tasks typical of traditional benchmarks, featuring a fixed goal and a sequential, linear execution path. **Middle:** LOGICWorld, where the execution path is governed by deep propositional logic and changes dynamically. This mandates the LLM agent to check environmental state in real-time and perform complex logical evaluations to dynamically determine the final goal. **Right:** Automatically generated Task Descriptions and Goal Verification functions.

To rigorously evaluate the reasoning frameworks required by these multi-branch tasks, we establish a robust system infrastructure for both automated logical task generation and stable agent execution (illustrated in Fig. 1). Because the environment is partially observable, the robot must actively interact with its surroundings to resolve logical predicates (e.g., verifying the contents of a closed cabinet) before deciding its next macro-action.

We conduct a comprehensive evaluation of state-of-the-art LLM-driven cognitive agents within LOGICWorld. By explicitly analyzing the models’ internal Chain-of-Thought (CoT) planning and execution trajectories, we quantify the operational efficiency and cognitive robustness of these “brains.” Our analysis reveals a stark reality: while modern agents are highly proficient at spatial execution, their capacity to maintain logical coherence collapses under deep propositional constraints, exposing a critical bottleneck in deploying them for real-world conditional tasks.

The contributions of this work are summarized as follows:

- We introduce a logic-centric robotic cognitive framework that embeds explicit propositional expressions into dynamic embodied instructions, moving beyond linear task execution.
- We develop a unified infrastructure for automated logical task generation and stable evaluation, specifically designed to test a robot’s logical state-machine tracking under partial observability.
- We provide a comprehensive evaluation and reasoning diagnostic of state-of-the-art LLM-driven embodied agents, revealing critical gaps in their dynamic replan-

ning and conditional branching capabilities.

II. RELATED WORK

A. Classical Planning Methods

The foundation of robust embodied action lies in automated planning. In the field of classical Artificial Intelligence planning, the Planning Domain Definition Language (PDDL) [10] remains the dominant knowledge representation language. PDDL formalizes environmental states and goals by defining the preconditions and effects of actions, allowing automated planners to search for a sequence of steps from an initial state to a desired goal state. As task complexity evolved, PDDL was extended: PDDL2.1 [11] introduced the capability to model time and continuous resources, enabling the system to handle metric constraints closer to real-world physical scenarios. For dealing with large-scale, structured problems, the Hierarchical Domain Definition Language (HDDL) [12] offers superior expressiveness by introducing task networks and abstract methods, allowing the agent to decompose complex problems at a high level.

Despite the formal rigor of these classical planning approaches, their reliance on a complete, accurate, and symbolic representation of the environment is a significant limitation. In embodied robotics, the perception step—translating raw sensory input into the precise symbolic state required by PDDL under partial observability—is a formidable challenge, restricting its direct application in open-world environments with visual uncertainty and ambiguity. LOGICWorld bridges this historical gap by effectively combining the strict, formal logical structure of PDDL-like constraints with

the open-world semantic perception and flexible reasoning capabilities of modern Large Language Models.

B. Embodied Benchmarks and Challenges

Early embodied agent benchmarks, such as ALFRED [6] and the underlying AI2-THOR environment [13], were philosophically influenced by classical planning. These benchmarks require agents to complete complex household tasks that semantically map to a sequence of actions derived from a formal plan. While challenging, they typically focus on static, preset goals and single-step action planning. To bridge the gap between classical planning and the perceptual world, many works have attempted to integrate symbolic planners with perception modules or utilized handcrafted finite state machines to guide agent behavior. However, these methods often struggle to adapt to dynamic environmental changes. For instance, when a task objective changes mid-execution due to newly discovered information (e.g., finding the target object is missing from its expected location), planners defined with static PDDL or HDDL face considerable difficulties in executing flexible conditional branching and immediate self-reflection.

The recognition of these limitations has spurred the development of more recent, sophisticated embodied benchmarks. A primary focus of these efforts has been on expanding environmental scale, diversity, and modality [8], [9], [14]–[16]. While these newer benchmarks excel at assessing perceptual robustness, generalization, and physical realism, the majority of their underlying task objectives still rely on sequentially executed goal lists or simple conjunctions of conditions. This persistent simplicity in task logic structure remains a major bottleneck for evaluating the dynamic state-machine tracking and conditional replanning capabilities required by real-world household robots. LOGICWorld addresses this core challenge by embedding structured, deep propositional constraints into the evaluation framework, forcing agents to actively explore and logically adapt.

C. LLM-based Cognitive Engines for Robotics

Inspired by the potent reasoning capabilities of Large Language Models (LLMs) and Vision-Language Models (VLMs), the focus in embodied agent research has rapidly shifted from low-level action planning to high-level cognitive reasoning [3], [17], [18]. The recent emergence of Large Reasoning Models (LRMs) such as OpenAI’s o1 [1] and DeepSeek-R1 [2] has demonstrated impressive long-sequence, step-by-step reasoning capabilities, offering promising solutions for complex decision-making problems.

Some studies have directly integrated language models as the central “brain” for embodied tasks [19]–[21]. For instance, frameworks like PaLM-E [22] and other approaches [23]–[25] utilize the inherent world knowledge of LLMs and combine it with robotic visual-motor data for training. Another body of research [26], [27] emphasizes the integration of LLMs into robotics by defining a high-level robot function library, prompting the LLM to compose functions

TABLE I
DETAILED TASK SUBTYPE DISTRIBUTION CATEGORIZED BY REQUIRED COGNITIVE CAPABILITIES.

Capability	Subtype	Action	Desc. Len.	# Preds	# Count
Spatial Execution	Basic	Navigate	9.40	1	10
		PickUp	8.25	1	20
		TurnOn/Off	8.50	1	20
		Place	7.02	1	50
State Tracking	Multi-Goal	Place	11.34	2	100
		Priority			
	Priority	Navigate	29.56	3	25
		PickUp	39.04	3	50
		Place	31.52	3	75
	Condition	Navigate	28.08	6	50
		PickUp	26.11	6	100
		Place	28.36	6	150
Comp. Reasoning	Logic	All	201.20	44	532
Total	-	-	-	-	1182

that achieve specified goals. However, despite these advancements, the capacity of these LLM-driven cognitive engines to accurately track internal logical states and handle multi-branch conditional constraints during long-horizon physical execution remains under-explored. LOGICWorld systematically fills this gap by providing a rigorous diagnostic testbed for these models’ logical robustness.

III. BENCHMARK SETUP: LOGICWORLD

A. Task Design: A Cognitive Capability Framework

Rather than simply scaling task length, LOGICWorld introduces a systematic progression of difficulty categorized by the specific cognitive and operational capabilities required by the agent (detailed in Table I). This framework systematically evaluates an agent’s transition from basic physical embodiment to complex, non-linear logical reasoning.

Spatial Execution & Long-Horizon Navigation. This category establishes the foundational baseline for embodied intelligence, focusing entirely on spatial and operational execution without imposing dynamic logical constraints. These tasks are designed to evaluate the agent’s core capabilities—such as search, interaction, and primitive skill chaining across the extended environmental scale of PROCTOR-10K. Success requires reliable, long-horizon sequential execution (e.g., exploring multiple rooms to locate and interact with specific objects) rather than conditional reasoning.

State Tracking & Deterministic Constraints. This category evaluates the agent’s working memory, state maintenance, and adherence to explicit rules. It introduces deterministic constraints that require structured planning and basic cognitive flexibility. Tasks in this category map to strict sequences, categorical assignments, or simple *if-else* conditional chains. These rigorously test the agent’s ability to maintain execution logic and adapt to immediate environmental feedback. The challenges are segmented into three distinct subtypes:

- 1) **Priority (Strict Sequential Execution):** Requires the agent to complete a series of subtasks in a strictly predefined order. This tests the agent’s ability to follow

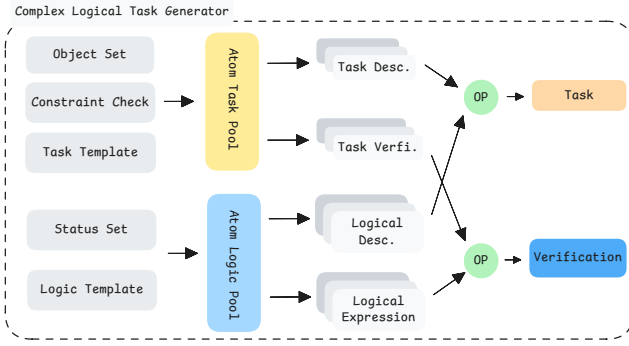


Fig. 2. Task Description and Verification Function Synthesis Pipeline.

rigid sequential constraints and maintain a memory of completed steps without deviating from the prescribed timeline.

- 2) **Conditional (Deterministic Branching):** Introduces basic decision-making based on environmental observations. The agent must evaluate explicit `if-else` conditions (e.g., “If object X is found, perform action Y; otherwise, perform action Z”) to determine the correct execution pathway. This tests its capacity for state-dependent plan adjustment.
- 3) **Multi-Goal (Cardinality & Categorization):** Involves handling multiple target instances or categorical assignments (e.g., placing several specific items or placing objects that belong to a broader semantic category). This challenges the agent’s continuous state tracking (e.g., managing the count of completed goals) and common-sense grounding without imposing strict sequential rules.

Compositional Reasoning & Dynamic Decision-Making. This category introduces the highest level of structural complexity, assessing the agent’s capacity for non-linear planning, dynamic state tracking, and multi-layered conditional execution. Tasks in this category are constructed as complex composite expressions involving 3–6 branches, where each branch contains 3–6 subtasks governed by compound logical expressions of 1–6 core predicates (e.g., $Exist(A)$, $Saw(A)$, $Visited(A)$). The agent must dynamically evaluate these Boolean expressions to implement conditional branching based on current environmental states and historical facts.

To construct these complex samples, as shown in Fig. 2, we employ a randomized logic expression generation mechanism composed of four structured stages:

- 1) **Atomic Task Construction:** The system samples a set of atomic tasks from a predefined pool and combines them using Boolean operators *AND*, *OR*. Corresponding validation functions are generated to ensure alignment between task representation and verification logic.
- 2) **Logical Condition Generation:** For each composite task, logical condition expressions and natural language descriptions are generated and incrementally

expanded using *AND*, *OR*, and *NOT*. To guarantee logical feasibility, a subset of conditions is constrained to always evaluate to `True`.

- 3) **Expression Composition:** Each condition is bound to the task’s verification function using the *AND* operator. The corresponding natural language instruction is formatted conditionally, enabling a direct mapping to executable constraints.
- 4) **Fallback Strategy Integration:** All conditional tasks are aggregated using the *OR* operator. To ensure completeness, a fallback branch is introduced: if no preceding conditions are satisfied, a final composite task is executed.

Ultimately, the generated output consists of a structured natural-language instruction and a corresponding Boolean verification expression used to determine execution pathways.

Mathematical Formulation. The synthesis of complex conditional tasks is formally defined by the global validation function T , which checks whether at least one conditional action branch satisfies both its logical requirements and execution criteria:

$$T = \bigvee_{k=1}^n [f_k(c_{j_1}, c_{j_2}, \dots; \mathcal{L}) \wedge g_k(V(t_a), V(t_b), \dots; \mathcal{L})] \quad (1)$$

where:

- T : The final verification result of the entire task.
- $\bigvee_{k=1}^n$: Generalized disjunction, indicating at least one branch must be satisfied.
- $f_k(\cdot)$: The Condition Composer function for branch k , evaluating a compound Boolean expression of logical state predicates (c_j).
- c_j : An individual logical predicate representing environmental facts (e.g., $Exist(A)$, $Saw(B)$).
- $g_k(\cdot)$: The Sequence Validator function for branch k , verifying the completion of atomic tasks via their verification functions $V(t)$.
- $V(t)$: The verification function for an atomic task t , returning `True` only when completed.
- $\mathcal{L} = \{\wedge, \vee, \neg\}$: The set of logical operators used for composition.

Scaling Task Complexity. To systematically calibrate the cognitive load in compositional reasoning, we parameterize task complexity through the monotonic scaling of three structural variables: the number of logical branches ($N_{branches}$), state predicates ($N_{predicates}$), and required subtasks ($N_{subtasks}$). The generation process scales progressively from a foundational baseline ($N_{branches} = 3$, $N_{predicates} \in [1, 2]$, $N_{subtasks} \in [3, 4]$) up to a maximum cognitive ceiling. At peak complexity, tasks demand the navigation of dense logical trees and deep execution sequences ($N_{branches} = 6$, $N_{predicates} \in [5, 6]$, $N_{subtasks} = 6$). This continuous parametric expansion ensures that the required planning depth increases monotonically, providing a rigorous stress test for LLM agents.

B. Environment: Leveraging PROCTOR-10K for Generalization

Simulator & Scene Diversity. LOGICWorld is built upon the AI2-THOR simulator [13] and grounded in the massive PROCTOR-10K dataset [28]. Unlike prior benchmarks restricted to small-scale scenes (e.g., 120 scenes in AI2-iTHOR or 89 in RoboTHOR), we leverage 1,000 diverse procedural environments featuring 16 distinct room layouts (ranging from 1 to 10 rooms) and 95 interactable object categories. This scale is critical for mitigating environment-specific overfitting and rigorously evaluating generalization.

Action Space & Abstraction. To focus evaluation on high-level compositional reasoning rather than low-level control, we abstract away fine-grained manipulation. The agent operates with a set of high-level skill primitives: *Navigate*, *PickUp*, *Open*, *Close*, *TurnOn*, *TurnOff*, *PutIn*, *PutOn*. This abstraction ensures that failures stem from logical planning errors rather than motor control noise.

Dynamic vs. Static Instructions. A key distinction of LOGICWorld is the dynamic nature of its directives. While benchmarks like ALFRED provide static, granular step-by-step instructions (e.g., “turn around and walk to the microwave”), LOGICWorld defines high-level goals modulated by latent environmental states and logical conditions. The agent cannot simply follow a pre-calculated path; it must proactively reason, self-reflect, and adapt its plan based on real-time observations.

Partial Observability and Active Exploration. Crucially, the environment operates under Partial Observability. The agent is not provided with ground-truth logical states (e.g., the system does not explicitly broadcast $\text{Exist}(\text{Apple in Fridge}) = \text{False}$ a priori). Instead, the agent must engage in active exploration using actions like $\text{Open}(\text{Fridge})$ and actively parsing the returned text-based or visual observation state to incrementally update its internal belief state. Only through this dynamic physical exploration can the agent resolve the logical predicates required by formula 1 to accurately determine the next execution branch.

C. LLM-driven Agent Formulation

The agent’s objective is to determine the optimal decision sequence that maximizes task completion probability given the instruction i and execution history. Formally, at time step t , the decision d_t is selected as:

$$d_t = \arg \max_{d \in \mathcal{D}} P(d \mid i, s_0, a_0, \dots, s_{t-1}) \quad (2)$$

where $\mathcal{D}$ represents the executable action space, s_0 is the initial state, and the history includes past observations and actions.

Reasoning-Action Loop with Action Chunking. Unlike traditional agents that plan one step at a time, our framework implements a CoT driven Action Chunking mechanism. In each interaction round, the LLM performs a four-stage process:

- 1) **Reflection:** Parses the latest feedback s_{t-1} to identify success or failure causes.

- 2) **State Update & Reasoning:** Updates internal belief state and deduces necessary logical preconditions.
- 3) **Planning:** Formulates a high-level natural language plan.
- 4) **Action Chunk Generation:** Generates a sequence of 1 to k primitive actions (e.g., *Navigate*; *Open*; *PickUp*) to be executed consecutively.

This chunking strategy significantly improves execution efficiency and temporal coherence. The environment executes the chunk sequentially and halts immediately upon any failure, returning the exact failure point and fresh observational feedback appending to the history for the next reasoning cycle. This mechanism is surprisingly robust because high-level LLMs tend to chunk deterministic primitive actions together while pausing for reflection before actions that require complex logical branching.

IV. EXPERIMENTS

A. Experimental Setup

Environment and Tasks. We conduct all experiments within the LOGICWorld benchmark, which is fundamentally built upon the PROCTOR-10K environment. Our comprehensive test set consists of 1182 tasks, systematically categorized by cognitive capability: *Spatial Execution*, *State Tracking*, and *Compositional Reasoning*. This structured categorization enables a granular assessment of LLM performance progression across different cognitive loads. All models are evaluated with a temperature of 0 to ensure reproducibility, and the maximum number of environment steps is 100 for all tasks.

Evaluated Models. We evaluate the interactive reasoning capabilities of several leading LLMs. Given the highly conditional and multi-turn nature of LOGICWorld, our evaluation focuses on comparing leading proprietary models (e.g., GPT-5-Mini, Claude-4.5-Haiku) and Qwen open-source series models. All models are integrated into our unified Agent framework, ensuring that performance differences are primarily attributable to the models’ core logical and reasoning capabilities, not disparate execution platforms.

Metrics. To quantify task performance and planning efficiency, we employ three standard metrics:

- **Success Rate (SR):** The primary metric, measuring the percentage of tasks successfully completed. A task is considered successful only if all logical conditions are satisfied and the final goal state is achieved within the maximum turn limit.
- **Success Weighted by Path Length (SPL):** To assess execution efficiency, SPL balances success rate against path length. It is defined as $\frac{1}{N} \sum_{i=1}^N S_i \frac{L_i}{\max(L_i, P_i)}$, where S_i is a binary indicator of success, L_i is the shortest path distance from the agent’s starting position to the goal, and P_i is the actual path length taken by the agent.
- **Average Actions Per Turn (Act/T):** This metric quantifies the density of productive decision-making within each interaction cycle. It is calculated as the total

TABLE II

PERFORMANCE COMPARISON ACROSS DIFFERENT COGNITIVE CAPABILITY CATEGORIES ON LOGICWORLD. WE REPORT SUCCESS RATE (SR %), SUCCESS WEIGHTED BY PATH LENGTH (SPL %), AND AVERAGE ACTIONS PER TURN (Act/T).

Model	Spatial Execution			State Tracking			Comp. Reasoning			Overall		
	SR(%)	SPL(%)	Act/T	SR(%)	SPL(%)	Act/T	SR(%)	SPL(%)	Act/T	SR(%)	SPL(%)	Act/T
<i>Proprietary Models</i>												
GPT-5-Mini	93.00	45.94	2.55	78.36	46.04	2.88	65.04	34.41	2.85	73.60	40.80	2.84
Claude-4.5-Haiku	100.00	71.89	1.34	86.00	59.70	1.65	75.75	42.53	1.46	82.85	53.04	1.54
Gemini-2.5-Flash	94.00	63.31	1.01	78.73	61.77	1.00	63.72	49.15	1.00	73.27	56.22	1.00
Qwen3-Max	94.00	67.38	1.25	82.18	61.10	1.42	73.12	45.34	1.60	79.10	55.61	1.46
<i>Open-Source Models</i>												
DeepSeek-V3.2-Exp	89.00	70.03	1.05	72.91	58.55	1.15	70.49	49.65	1.08	73.18	55.52	1.11
Qwen3-14B	79.00	56.10	1.28	45.82	34.87	1.23	33.27	28.1	1.54	42.98	31.18	1.38
Qwen3-8B	75.00	59.08	1.03	29.09	24.29	1.16	30.08	25.4	1.00	33.42	25.55	1.05
Qwen3-4B	56.00	38.06	1.07	22.36	17.59	1.00	15.23	12.1	1.35	22.00	14.94	1.15

number of environment-changing actions divided by the number of turns taken. A higher Act/T, when coupled with a high SR, indicates superior long-horizon planning and the ability to compose complex macro-actions within a single reasoning step.

B. Main Results and Analysis

Table II presents the performance comparison of various proprietary and open-source models across three cognitive capability categories. Our analysis reveals several key findings regarding the current state of embodied interactive reasoning.

Cognitive Complexity Gap. A consistent performance degradation is observed across all models as task complexity scales from *Spatial Execution* to *Compositional Reasoning*. For instance, Claude-4.5-Haiku achieves a perfect Success Rate (SR) of 100.0% in spatial tasks, but its performance drops to 75.75% in the compositional reasoning tier. This trend confirms that the primary challenge in LOGICWorld lies in the non-linear logical branching and deep propositional constraints rather than low-level motor control or simple navigation.

Aggressive Planning vs. Execution Accuracy. The *Average Actions per Turn (Act/T)* metric provides a unique lens into the models' planning strategies.

- **Aggressive Chunking:** GPT-5-Mini demonstrates the highest planning confidence, with an average of 2.84 actions per turn. However, this aggressive "action-chunk" strategy does not translate to the highest success rate, particularly in State Tracking where its SPL (46.04%) lags significantly behind Claude-4.5-Haiku (59.70%). This suggests that while GPT-5-mini can generate long execution sequences, it is more susceptible to error propagation in logically dynamic environments.
- **Conservative Precision:** In contrast, models like Gemini-2.5-Flash and DeepSeek-V3.2-Exp maintain a more conservative execution style ($\text{Act/T} \approx 1.0$ -1.1), prioritizing step-by-step verification which leads to higher path efficiency (SPL) in complex scenarios.

TABLE III

DETAILED SUCCESS RATE (SR %) ACROSS TASK TYPES.

Model	Basic	Condition	Multi-Goal	Priority	Logic
Claude-4.5-Haiku	100.0	88.0	85.0	82.7	76.3
Qwen3-Max	94.0	86.3	81.0	74.7	73.1
GPT-5-mini	93.0	81.0	78.0	73.3	65.0
Gemini-2.5-Flash	94.0	82.3	79.0	71.3	63.7
DeepSeek-V3.2-Exp	89.0	84.0	47.0	68.0	70.5
Qwen3-14B	79.0	53.0	32.0	40.7	33.3
Qwen3-8B	75.0	39.0	27.0	10.7	30.1
Qwen3-4B	56.0	32.0	17.0	6.7	15.2

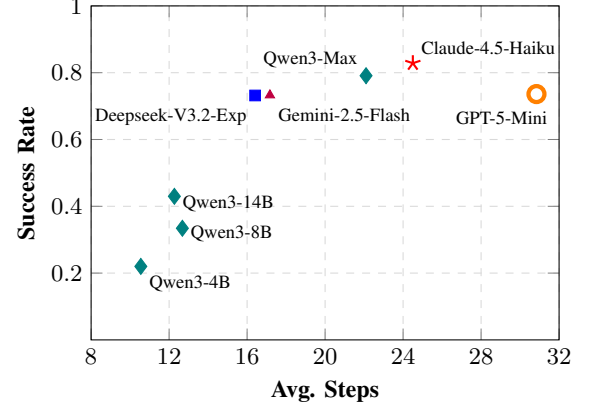


Fig. 3. Success Rate versus Average Execution Steps across various models. The plot illustrates the trade-off between task performance and planning parsimony.

Table III provides a heatmap of SR across five specific task archetypes, revealing where different model architectures struggle. Most models, including *Qwen3-Max*, show a steep decline when transitioning from *Condition* to *Priority* and *Logic*. The *Priority* subtype requires the agent to dynamically re-order sub-tasks based on environment constraints, a skill that remains a differentiator between frontier proprietary models and the open-source *Qwen3-14B/8B* series.

The Efficiency-Effectiveness Frontier. Fig. 3 illustrates the trade-off between task success and planning parsimony.

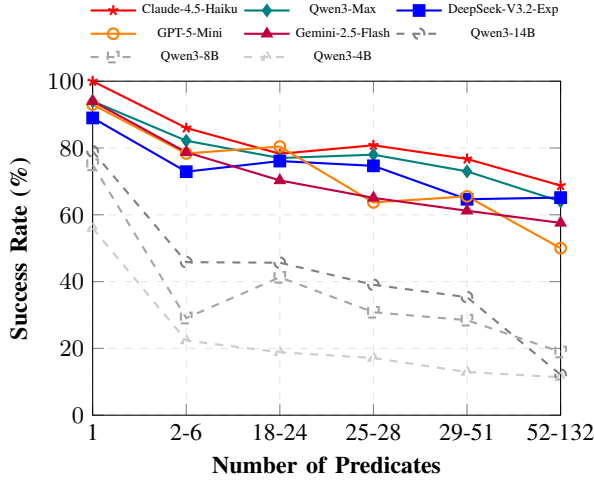


Fig. 4. Success Rate degradation as task complexity increases. Top-tier models maintain higher robustness, while smaller models exhibit sharp performance drops at higher complexity levels.

TABLE IV
IMPACT OF ACTION-CHUNKING ON SR, SPL, ACT/T AND TOKEN CONSUMPTION ON LOGIC TASKS.

Action-Chunking	SR(%)	SPL(%)	Act/T	Tokens
Enabled	75.75	42.53	1.46	39,764
Disabled	77.26	45.00	1.0	46,388

Claude-4.5-Haiku and *Qwen3-Max* occupy the upper-right frontier, achieving high success rates within reasonable step counts (22–25 steps). Notably, *DeepSeek-V3.2-Exp* and *Gemini-2.5-Flash* exhibit higher efficiency (17 steps) while maintaining competitive SR (73.27%), suggesting a more direct path-finding capability in embodied navigation. *GPT-5-mini* presents an anomalous profile; it requires the highest average steps (30.84) to achieve a moderate SR (73.60%). This confirms our earlier observation that it outputs multiple actions at once in a single turn.

Robustness to Structural Complexity. Fig 4 quantifies the relationship between task performance and the cardinality of propositional predicates, while *Qwen3-4B* and *8B* exhibit marginal success rates ($SR < 20\%$) as the predicate count exceeds 6, *Claude-4.5-Haiku* maintains a robust SR of 68.75% in the most complex regime (52–132 predicates). A significant performance attenuation is observed across most architectures when transitioning from the 29–51 to the 52–132 interval, with the exception of frontier proprietary models. This trend suggests that maintaining global plan coherence in environments with high logical entropy remains a primary differentiator of model reasoning depth.

C. Action-Chunking Ablation

The primary objective of Action-Chunking is to enhance operational efficiency by reducing the number of costly LLM calls. We conduct an ablation study using *Claude-4.5-Haiku*, comparing the Action-Chunking enabled mode against the disabled mode.

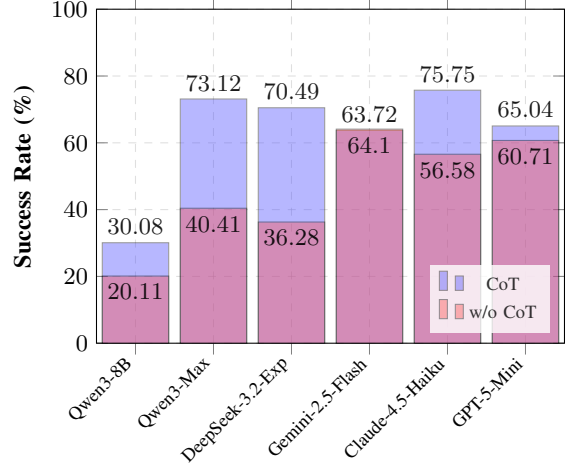


Fig. 5. Comparison of success rates with and without CoT across different models.

Table IV illustrates the performance trade-offs inherent in the Action-Chunking mechanism. The empirical results indicate that enabling Action-Chunking incurs only a negligible attenuation in Success Rate (-1.51%) and path efficiency (-2.47% SPL). This stability suggests that frontier models like *Claude-4.5-Haiku* possess sufficient *temporal reasoning depth* to pre-calculate short-term action sequences without significantly compromising execution integrity.

Beyond raw performance, the primary advantage of the Action-Chunking mechanism lies in its superior resource optimization. By allowing the agent to consolidate multiple primitive actions into a single reasoning cycle, the *planning density* (Act/T) increases by 46% (from 1.00 to 1.46). This shift in interaction modality leads to a significant reduction in cumulative inference overhead, with total token consumption decreasing by 14.28%. These findings validate Action-Chunking as a robust efficiency-enhancing strategy, effectively amortizing the high latency and fiscal costs of LLM calls in long-horizon embodied tasks.

D. Reasoning Mechanism Ablation

The deep propositional constraints inherent in *Compositional Reasoning* necessitate explicit, structured inference. We conduct an ablation study to evaluate the functional necessity of the CoT mechanism against a direct action-generation baseline within this capability category.

CoT Dependency in Model Reasoning. As illustrated in Fig. 5, explicit reasoning is a critical prerequisite for success in the majority of evaluated models. For instance, *Qwen3-Max* and *DeepSeek-V3.2-Exp* exhibit a substantial performance degradation, with Success Rates (SR) dropping by 32.71 and 34.21 percentage points, respectively. This sensitivity indicates a *computational dependency* on the intermediate tokens provided by CoT, which serve as an externalized working memory to resolve state-tracking and multi-step logical branching.

Emergence of Implicit Reasoning Robustness. Conversely, the ablation reveals a divergent reasoning paradigm

in frontier models such as *Gemini-2.5-Flash* and *GPT-5-Mini*. The SR of *Gemini-2.5-Flash* remains statistically invariant, while *GPT-5-Mini* shows only a marginal 4.33% attenuation. This phenomenon suggests that these architectures possess superior *implicit reasoning capacity*; for these models, compositional constraints can be effectively internalized during the latent pre-computation of the action sequence, rather than requiring serialized verbalization. This highlights a significant dimension of model maturity where logical coherence in *Compositional Reasoning* is decoupled from explicit output length.

V. CONCLUSIONS

We introduced LOGICWorld, a benchmark designed to assess the capability of LLM-driven embodied agents by automatically generating dynamic instructions grounded in deep propositional logic. By enforcing a unified logic validation structure akin to Disjunctive Normal Form (DNF), LOGICWorld bridges the rigor of classical planning algorithms with the semantic generalizability of language models. Our experiments reveal that while action chunking and CoT prompting boost efficiency and operational working memory, even advanced models exhibit substantial degradation when faced with non-linear branching, deep logical dependencies, and long-horizon active exploration.

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